

Respiratory Failure with Protected Intubation

Section 1: Case Summary

Scenario Title:	Respiratory Failure with Protected Intubation
Keywords:	COVID-19, intubation, communication
Brief Description of Case:	A 70-year old woman who is known COVID-positive presents with hypoxic respiratory failure and requires urgent intubation. This needs to be done in a negative pressure room with airborne PPE. Communication between the team members inside and outside the room must be maintained throughout the clinical care.

Goals and Objectives	
Educational Goal:	Practice a protected intubation according to local protocols, with attention to personal and team safety, effective team communication and safe intubation.
Objectives: (Medical and CRM)	1. Organize the healthcare team by using a pre-procedure briefing 2. Maintain effective team communication throughout the procedure, including the team members outside the negative pressure room 3. Minimize personal risk by utilizing the appropriate donning/doffing of PPE 4. Perform high acuity clinical care (intubation) using local guidelines
EPAs Assessed:	None

Learners, Setting and Personnel			
Target Learners:	☒ Junior Learners	☒ Senior Learners	☐ Staff
	☐ Physicians	☐ Nurses	☐ RTs
	☐ Other Learners:		
Location:	☒ Sim Lab	☐ In Situ	☐ Other:
Recommended Number of Facilitators:	Instructors: 1		
	Confederates: 2		
	Sim Techs: 1		

Scenario Development	
Date of Development:	April 15, 2020
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Revised By:	
Version Number:	



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Section 2A: Initial Patient Information

A. Patient Chart					
Patient Name: Mary Francis		Age: 70		Gender: F	Weight: 70kg
Presenting complaint: Short of breath					
Temp: 38.1	HR: 95	BP: 110/70	RR: 30	O ₂ Sat: 81%	FiO ₂ : 100%
Cap glucose: 6.1			GCS: 15 (E V M)		
Triage note: Patient coming from home where she is isolating. Known COVID+. Non-productive cough for six days. Now increasing short of breath for two days. EMS found patient tachypneic with an SpO ₂ in the 70s on room air.					
Allergies: NKDA					
Past Medical History: Hypertension			Current Medications: Ramipril 5mg daily		

Section 2B: Extra Patient Information

A. Further History	
<i>Include any relevant history not included in triage note above. What information will only be given to learners if they ask? Who will provide this information (mannequin's voice, confederate, SP, etc.)?</i>	
HPI & PMHx: As above triage note. Social History: Patient independent at home, recently retired office worker, FULL CODE.	
B. Physical Exam	
<i>List any pertinent positive and negative findings</i>	
Cardio: Nil	Neuro: Nil
Resp: 2-3 work dyspnea, coarse bilateral crackles	Head & Neck: Nil
Abdo: Nil	MSK/skin: Nil
Other: Nil	

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Section 3: Technical Requirements/Room Vision

A. Patient
☒ Mannequin (Adult)
☐ Standardized Patient
☐ Task Trainer
☐ Hybrid
B. Special Equipment Required
Intubation supplies Telecommunication devices (speakerphone, walkie-talkie, baby monitor, etc) Ideally, a physical barrier or second room to use so that the “inside team” and “outside team” are separated and must use a telecommunication device to communicate.
C. Required Medications
RSI kit (COVID-specific)
D. Moulage
Elderly woman
E. Monitors at Case Onset
☐ Patient on monitor with vitals displayed
☒ Patient not yet on monitor
F. Patient Reactions and Exam
<i>Include any relevant physical exam findings that require mannequin programming or cues from patient (e.g. – abnormal breath sounds, moaning when RUQ palpated, etc.) May be helpful to frame in ABCDE format.</i> Breathing – Coarse Crackles throughout all lung fields

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Section 4: Confederates and Standardized Patients

Confederate and Standardized Patient Roles and Scripts	
Role	Description of role, expected behavior, and key moments to intervene/prompt learners. Include any script required (including conveying patient information if patient is unable)
RT (inside room)	Prepares intubation equipment and ventilator. Assists during intubation. Can be more or less hands-on depending on the training level of the participants.
Charting RN (outside room)	Stays outside the room, recording pertinent information. Uses telecommunication device to gather the information.

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Section 5: Scenario Progression

Scenario States, Modifiers and Triggers				
Patient State/Vitals	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State		Facilitator Notes
1. Baseline State EMS patches in with above triage note		<u>Expected Learner Actions</u> ☐ Recognize need for intubation ☐ Assign team roles ☐ Gather medications ☐ Gather supplies ☐ Pre-procedure briefing ☐ Discuss rescue plans	<u>Modifiers</u> <u>Triggers</u> 5 min → Patient arrival	
2. Patient Arrival Rhythm: Sinus HR: 95 BP: 110/70 RR: 30 O ₂ SAT: 81% (NRB) T: 38.1°C GCS: 15	Tachypneic Dyspneic	<u>Expected Learner Actions</u> ☐ Don PPE then enter room ☐ Check communication device ☐ Confirm code status ☐ Brief history/physical exam ☐ Secure IV access ☐ Prepare ventilator ☐ Team member to call ICU ☐ Preoxygenate ☐ Consider proning	<u>Modifiers</u> <u>Triggers</u> All action complete → Intubation	
3. Intubation O ₂ SAT: 81% (NRB) GCS: 3 (after intubation meds given)	Paralyzed	<u>Expected Learner Actions</u> ☐ Pre-intubation time out to review plan ☐ Push RSI meds ☐ Wait 60 seconds ☐ Intubation	<u>Modifiers</u> RSI meds given: SpO ₂ drops to 70% <u>Triggers</u> Passed tube → Post-intubation	
4. Post-Intubation RR: 12 O ₂ SAT: 90% (vent)		<u>Expected Learner Actions</u> ☐ Confirm tube placement ☐ Safely attach ventilator ☐ Provide sedation ☐ Doff PPE ☐ Call ICU (if not yet done)	<u>Modifiers</u> <u>Triggers</u> Doff PPE → End case	



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Appendix A: Laboratory Results

No labs given during case

Appendix B: ECGs, X-rays, Ultrasounds and Pictures

No imaging given during case

Appendix C: Facilitator Cheat Sheet & Debriefing Tips

Early in the COVID-19 pandemic, new protocols and procedures were developed to maximize provider safety during high risk clinical activities such as intubation and cardiac arrest resuscitation. Initially, learners were excluded from performing these procedures but as time went on and this restriction was relaxed, we realized that they needed practice before following these new protocols for actual patients.

We re-arranged our sim lab to more closely reflect the challenges faced in the actual emergency department. Some of the adjustments we made to our lab:

- Stripped out all equipment and made participants think about what they needed to take in
- Narrowed the medications available for RSI and sedation
- Kept the sim tech and charting nurse outside of the room and used a speakerphone or walkie-talkies to communicate between the inside and outside of the negative pressure room
- Used a white noise machine to simulate a HEPA filter
- Used expired or washable PPE (especially N95 masks) to simulate the communication barrier they create
- Assigned a confederate to inside the room (RT) and outside the room (charting RN) to get feedback from both sides of the wall

Some Sample Questions for Debrief:

- What were some barriers to effective communication? How were these overcome (or could be overcome in the future)?
- What information did you include in the pre-brief? Was anything missed? Did you use a checklist or tool, and was it helpful?
- How was this intubation different from your typical intubations? How did those differences make you feel?
- What strategies did you use to ensure safe donning and doffing of PPE?
- Discuss various non-invasive ways to improve oxygenation. Which of these are allowed at your institution?

*Below is one infographic highlighting key changes from usual intubation from the FOAM website CanadiEM. Note that as understanding of COVID grows, protocols will change to incorporate best practices and older resources may be obsolete.

**Use your most up to date local Protected Intubation protocol.



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Source: <https://canadiem.org/recommendations-for-covid-19-intubation-an-infographic/>

References

1. <https://emergencymedicinescases.com/covid-19-protected-intubation/>
2. <https://empendium.com/mcmtextbook/covid-19/234191.covid-19-mcmaster-survival-guide>
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